3951. Minimum Energy to Maintain Brightness

Solved

Medium

Hint

You are given an integer `n`, representing `n` light bulbs arranged in a line and indexed from 0 to `n - 1`.

You are also given an integer `brightness` and a 2D integer array `intervals`, where `intervals[i] = [start_i, end_i]` represents an **inclusive** time interval during which the lighting requirement **must** be satisfied.

At each time unit, every bulb can independently be either on or off. A bulb that is on **illuminates** its own position and its **adjacent** positions, if they exist.

The **total illumination** at a time unit is the number of **illuminated** positions. Each position is counted **at most once**.

For every integer time unit covered by **at least** one interval in `intervals`, the **total illumination** must be **at least** `brightness`. At time units not covered by any interval, all bulbs may remain off. Each bulb that is on consumes 1 unit of energy for that time unit.

Return an integer denoting the **minimum** total energy required.

Example 1:

> **Input:** n = 5, brightness = 5, intervals = [[6,12]]
>
> **Output:** 14
>
> **Explanation:**
>
> - Turn on the light bulbs at positions 1 and 4.
> - Current state of line: `0 1 0 0 1`.
> - All 5 positions are illuminated, so the required brightness is reached.
> - The active interval has length `12 - 6 + 1 = 7`, so the total energy is `2 * 7 = 14`.

Example 2:

> **Input:** n = 2, brightness = 1, intervals = [[0,0],[2,2]]
>
> **Output:** 2
>
> **Explanation:**
>
> - Turn on one light bulb during each active interval.
> - Each interval has length 1, so the total active time is `1 + 1 = 2`.
> - The total energy is `1 * 2 = 2`.

Example 3:

> **Input:** n = 4, brightness = 2, intervals = [[1,3],[2,4]]
>
> **Output:** 4
>
> **Explanation:**
>
> - Turn on one light bulb. It can illuminate at least 2 positions.
> - The active intervals overlap, so the total active time is the length of `[1,4]`, which is 4.
> - The total energy is `1 * 4 = 4`.

Constraints:

- $1 <= n <= 10^6$
- `1 <= brightness <= n`
- $1 <= intervals.length <= 10^5$
- $intervals[i] == [start_i, end_i]$
- $0 <= start_i <= end_i <= 10^9$

Seen this question in a real interview before? 1/6

Yes No

Accepted 21,159 /39K | **Acceptance Rate** 54.2 %

Topics

Hint 1

Hint 2

Hint 3

Hint 4

Discussion (26)